VEPOLINK
MONITORING PURITY

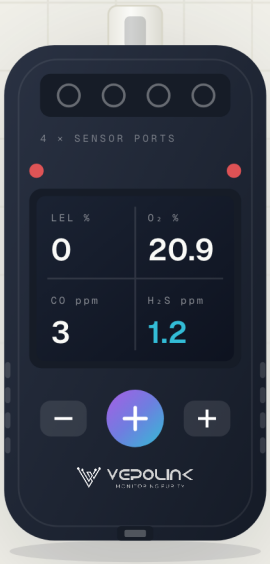
ENVIRONMENTAL INTELLIGENCE × GAS SAFETY

PORTABLE GAS DETECTION · CONFINED SPACE

Portable Multi Gas Detector

Four gases, one hand

Combustibles, oxygen and two toxic gases read continuously and simultaneously on one colour display — the standard kit for confined-space entry, mining and anywhere the atmosphere is unknown.



4 gases
SIMULTANEOUS CHANNELS

< 30 s
RESPONSE TIME T90

1200 rec
ON-BOARD DATA RECORDS

4-GAS LEL · O_2 · CO · H_2S | **32-BIT** COLOUR LCD | **IP** RUBBER-ARMOURED

4	0–30%	<30s	1200
GASES READ SIMULTANEOUSLY	OXYGEN MEASURING RANGE	RESPONSE TIME T90	LOGGED RECORDS VIA USB

01 OVERVIEW

The whole atmosphere, *at a glance.*

A multi-gas detector reads several gases with one instrument. The Vepolink four-gas unit monitors combustible gas, oxygen and two toxic gases continuously and simultaneously — the combination confined-space and mining work demands, where the hazard is not one known gas but whatever the atmosphere turns out to hold.

A 32-bit MCU drives a colour LCD showing all four channels at once, with self-adapting sensor management and 1200 on-board records that transfer to a PC for printing and analysis. The housing is engineering plastic over anti-slip rubber: waterproof, dustproof and rated for explosion-hazard areas.

STANDARD CHANNELS

- COMBUSTIBLE % LEL
- OXYGEN % VOL
- CO PPM
- H_2S PPM
- TOXIC SWAP OPTIONS
- DIFFUSION
- PUMPED OPTION
- USB OUTPUT

· PRODUCT FEATURES

Four Channels, One Screen

A colour LCD shows LEL, O_2 and both toxic readings together, so an unsafe combination is obvious at a glance rather than pieced together from four instruments.

32-bit Self-Adapting MCU

The controller manages sensor drift and temperature behaviour automatically, holding accuracy across a shift without repeated manual adjustment.

Armoured Housing

High-strength engineering plastic with a compound anti-slip rubber jacket — waterproof, dustproof and explosion-protected for underground and process areas.

1200-Record Logger

Readings and alarm events are stored on board and downloaded over USB to PC software for printing, exposure records and analysis.

02 SENSING PRINCIPLES

Four ways to *see a gas.*

Four channels need more than one kind of sensor. The instrument combines the technologies below so each gas is measured the way its chemistry allows.

ELECTROCHEMICAL

Toxic Gases & Oxygen

CHANNELS 2, 3 & 4 — O_2 · CO · H_2S



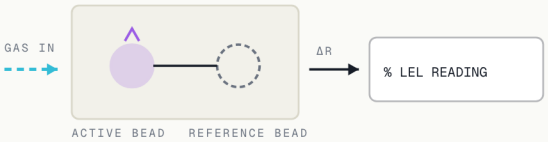
Gas diffusing into the cell is oxidised or reduced at the sensing electrode, producing a current in direct proportion to concentration. Low power, ppm resolution, and selective by electrode chemistry.

`Current ∝ concentration`

CATALYTIC BEAD

Combustible Gases

CHANNEL 1 — CH_4 · LPG · H_2 · HEXANE



Flammable gas burns on a heated catalytic bead, raising its resistance relative to an inert reference bead. Extended response covers pentane, hexane and other long-chain hydrocarbons alongside methane.

`Reads in % LEL`

NDIR

Carbon Dioxide & Hydrocarbons

OPTIONAL TOXIC-SLOT ALTERNATIVE



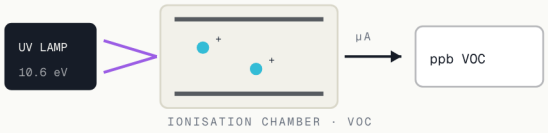
An infrared beam crosses a gas cell at a wavelength the target absorbs. The drop in light reaching the detector gives concentration — with no oxygen needed and no poisoning of the sensor.

`Absorption ∝ concentration`

PID

Volatile Organic Compounds

OPTIONAL TOXIC-SLOT ALTERNATIVE



Ultraviolet light ionises organic vapours passing between two electrodes; the resulting ion current is measured. Sensitive to the ppb level for solvents, fuels and many industrial VOCs.

`VOC screening`

03 CHANNEL CONFIGURATION

Four slots. *Your combination.*

The standard build is combustibles, oxygen, carbon monoxide and hydrogen sulphide. The two toxic slots can be configured for the gases below to match your process.

O_2	Oxygen STD	CO	Carbon Monoxide STD	H_2S	Hydrogen Sulphide STD	SO_2	Sulphur Dioxide EC
NH_3	Ammonia EC	Cl_2	Chlorine EC	HCl	Hydrogen Chloride EC	HCN	Hydrogen Cyanide EC
NO	Nitric Oxide EC	NO_2	Nitrogen Dioxide EC	PH_3	Phosphine EC	$COCl_2$	Phosgene EC
O_3	Ozone EC	H_2	Hydrogen CAT	LPG	LPG / Propane CAT	LEL	Combustibles STD
C_6H_{14}	Hexane CAT	C_2H_4	Ethylene CAT	CO_2	Carbon Dioxide NDIR	VOC	Volatile Organics PID

STEP 01

Bump and go

Power on, watch all four channels self-test, confirm response to a quad test gas, then clip it on.

STEP 02

Sample

Natural diffusion through the sensor ports, or the auto-suction pump and probe to test a tank, pit or manhole before entry.

STEP 03

Alarm

Any channel crossing its threshold fires buzzer, LEDs and vibration, with the offending gas highlighted on the display.

STEP 04

Log and review

Up to 1200 records are stored on board and transferred to PC software for printing and exposure analysis.

DIFFUSION OR PUMPED

In diffusion mode the sensors breathe the air around the wearer — the right choice for personal protection. The auto-suction option adds a pump and sample hose so a confined space can be tested top to bottom from outside before anyone enters, as entry permits require.

SETTING ALARM LEVELS

Each channel carries its own thresholds: typically 10 % and 20 % LEL for combustibles, 19.5 % and 23.5 % for oxygen, and TWA / STEL values for CO and H_2S. All are retained through power cycles and sensor changes.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

Channels	4 — simultaneous, continuous
Monitoring Parameters	Any combustible gas, toxic gases and oxygen
Standard Configuration	% LEL · O_2 · CO · H_2S
Oxygen Range	0 – 30 % vol
Resolution	0.1 %
Accuracy	< ±2 % of full scale
Response Time	T90 < 30 s
Sampling Type	Diffusion · auto-suction pump optional
Sensor Type	High-sensitivity, high-repeatability sensors
Calibration	Zero and span with standard calibration gas

ALARM · POWER · BUILD

Display	Colour LCD, real-time status of all channels
Controller	32-bit MCU, self-adapting
Alarm Indication	Buzzer · LED · vibration, per channel
Data Records	1200 on board · PC software for print and analysis
Digital Output	USB
Operating Power	Rechargeable battery
Charger	230 VAC, 50 Hz
Housing	Engineering plastic + anti-slip rubber · waterproof, dustproof, explosion-protected
Weight	Less than 300 g

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



WEIGHT
< 300 g

05 USE & UPKEEP

Four sensors, *one routine.*

A multi-gas monitor is only as good as its last check. Good practice is a **four-gas bump test before every shift**, a **full calibration at scheduled intervals** with certified quad gas, and a sensor change when any channel slows — all of which the user can do without sending the unit away.

01 · WEARING

In the breathing zone

Clip to a collar, lapel or harness so the sensors sample the air the wearer breathes, ports unobstructed.

02 · BUMP TEST

Daily, seconds

A short exposure to quad test gas confirms all four channels and the alarms respond before the unit enters a hazardous area.

03 · CALIBRATION

Zero and span

Zero all channels in clean air, span with a certified four-gas mix. Remote guidance from Vepolink engineers is available for the first few cycles.

04 · CHARGING

Overnight, shift-ready

Return to the charger at shift end. A low-battery warning sounds well before shutdown, so the unit never goes quiet unnoticed.

05 · SENSOR CHANGE

User serviceable

Each sensor swaps out individually in minutes. Recalibrate the affected channel and the unit is back in service.

06 · RECORDS

Downloaded over USB

Up to 1200 logged records export to PC software for printing, exposure history, audits and incident investigation.

MULTI GAS OR SINGLE GAS?

Where the atmosphere is uncertain — a confined space, a sewer, an underground working, a vessel after cleaning — four channels in one housing are the only sensible choice, and most entry permits require exactly this set. Where the hazard is a single known gas worn all day, the Vepolink portable single-gas detector is lighter and simpler to keep.

FLEET SUPPLY

Units are supplied individually or as a site fleet with quad calibration gas, regulators, sampling pumps and hoses, charging cradles and a spares kit of sensors and filters, so the safety team can keep every detector in service from its own store.

06 APPLICATIONS

Where it *works.*

Anywhere the atmosphere is unknown before someone goes in — and anywhere explosion protection is a condition of entry.

Refineries & Petrochemicals	Chemical & Fertiliser Plants	Confined-Space Entry Permits
Sewers & Water Treatment	Power & Boiler Houses	Steel & Metallurgy
Oil Rigs & Offshore	Tunnels & Shipyards	Mining & Underground Channels

PERSONAL AND PROCESS PROTECTION

The detector serves two audiences: the entrant, who gets an immediate warning to leave, and the site, which gets a logged record of every pre-entry test, exposure and alarm event. Paired with fixed detectors and a control interface, a portable unit closes the gap in the places fixed heads cannot reach — inside vessels, under floors, at the end of a hose.

CHOOSING THE TOXIC SLOTS

CO and H_2S cover most confined-space work. Where the process demands it the two toxic slots can carry NH_3 for refrigeration, Cl_2 or SO_2 for water and chemical plants, or NO_2 for diesel exhaust underground. Tell us the process and we will recommend the combination and alarm levels.

07 FAQ

Frequently asked *questions.*

Q What does the multi-gas detector do?

It is a handheld safety detector that monitors combustible gases, toxic gases and oxygen simultaneously — four channels on one display — for industrial safety.

Q Which gases can it measure?

Combustible gases in % LEL, oxygen, CO and H_2S as standard, with the two toxic slots configurable for other toxic gases to suit hazardous environments.

Q Is it suitable for industrial safety monitoring?

Yes. It is designed for industrial safety, confined-space entry monitoring and gas-leak detection where explosion protection is required.

Q What are the key features?

High-sensitivity sensors, a real-time colour LCD, a 32-bit self-adapting MCU, a compact armoured housing and 1200-record data storage with USB transfer for on-site analysis.

Q Does it support data logging?

Yes. Up to 1200 records are stored on board and transferred to a computer over USB for printing and detailed exposure analysis.

Q What is the sampling method?

Both natural diffusion and auto-suction are offered. Diffusion suits personal wear; the pumped version draws a sample through a hose for pre-entry checks.

Q How are alarms set?

Each channel has its own thresholds, retained in memory. Buzzer, LED and vibration fire together, with the alarming gas highlighted on screen.

Q Can the user calibrate it?

Yes. With standard four-gas calibration mix, zero and span calibration is done on site — with remote help from Vepolink engineers if wanted.

Q Can it interface with a control system?

Portable units alarm locally. For automatic process shutdown, pair them with Vepolink fixed gas detectors wired to the plant control system.

Q Is the housing rugged?

Yes. High-strength engineering plastic with a compound anti-slip rubber jacket — waterproof, dustproof and explosion-protected.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. Gas detectors are safety equipment: follow the user manual, site procedures and applicable standards for bump testing, calibration and use. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

FOUR GASES, WATCHED TOGETHER

Know the atmosphere *before you enter.*

Tell us the process, the spaces to be entered and how many people need protecting, and we'll configure the channel set, alarm levels, quad calibration gas, pumps and charging kit as one supply. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specifies, calibrates and supports gas detection. No call centre.
+91 99538 86178 →

02 **Request a fleet quotation** RECOMMENDED
Send your process and headcount — we'll spec channels, alarm levels, quad gas, pumps, chargers and spares.
sales@vepolink.com →

03 **Book a demonstration**
See a four-gas bump test, a pumped pre-entry check and a calibration with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM